

Phenotype Transfer Maps

Cross-study organoid cell typing with measured sample review priorities

Yan Su

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Team: Yan Su (individual entrant)

Measured result: A Phenotype Transfer Map joins a cross-acquisition-source phenotype score, score-based review order within an atlas-selected cohort, and separate label-backed error audit. A fixed Velasco-trained model reached **0.9078 macro-F1** on **207,871 Bhaduri neural-organoid cells** (34 sample-key bootstrap interval **0.8423-0.9372**). It beat a source-trained nearest-centroid comparator by **+0.1010** on the same cells (paired interval **+0.0705 to +0.1205**). HNOCA's gene panel and labels were harmonized across studies; this is not an independent biological ground truth.

Exploratory review queue: Ranking the 34 Bhaduri sample keys by mean model-score uncertainty, without using their labels in the ranking calculation, correlated with their observed classification error (Spearman **0.7791**, key-resampling interval **0.5546-0.9010**). The highest-ranked eight keys had **1.721 times** the equal-key average error rate; a post hoc, fixed-rank shuffle of their 34 observed error rates reached this ratio 0/10,000 times (plus-one **p=0.00010**). HNOCA labels selected the three-class cohort before ranking. This is retrospective evidence on the same test source, not an end-to-end unannotated-batch validation or calibrated error probability.

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1 Abstract

A Phenotype Transfer Map measures a cell-type model across acquisition sources, orders atlas-selected sample keys by model-score uncertainty for review, and separately audits errors where reference labels exist. We trained on 135,053 cells from Velasco et al. and evaluated once on all 207,871 selected cells from Bhaduri et al. in the public Human Neural Organoid Cell Atlas (HNOCA) [9-12]. The held-out-study macro-F1 was **0.907831**; a 2,000-draw bootstrap over 34 `bio_sample` keys gave a 95% interval of **0.842324-0.937195**. A source-trained nearest-centroid comparator scored 0.806783 on the same cells. The paired macro-F1 difference was +0.101048, with a sample-key interval of +0.070494 to +0.120536. Glioblast precision was only 0.679004: 1,353 cells of other classes were flagged as Glioblast. The usable action is to prioritize sample-level annotation review, not to treat predicted proportions as a new biological ground truth.

In a subsequent exploratory analysis, a queue based only on classifier decision-score margins ranked all 34 Bhaduri sample keys without using their HNOCA labels in the ranking calculation. Mean score uncertainty and observed key error rate had equal-key Spearman correlation **0.779068**, with a 2,000-draw key-resampling interval **0.554570-0.901040**. The eight most uncertain keys had **1.720951 times** the all-key mean error rate. After seeing this ratio, we kept those eight ranks fixed and shuffled the 34 observed group error rates 10,000 times; zero shuffles reached the observed ratio (one-sided plus-one **p=0.00009999**). HNOCA labels selected the three-class cohort before ranking, and the utility check reused the same Bhaduri source and labels; it has not been tested prospectively on a wholly unannotated batch.

A secondary annotation-source check matched Bhaduri’s preserved original labels to these cells. On 175,160 unambiguous Neuron/Radial Glia cells, binary macro-F1 was 0.941193 (34-key interval 0.922403-0.956609), but the fixed nearest-centroid comparator scored 0.947220; the paired model-minus-comparator interval was -0.010067 to -0.002531. This result does not add model superiority. It shows that the preferred model changes when the annotation source and endpoint change, and it is not a new laboratory or independently blind label set.

The two experiments were collected in different laboratories, but HNOCA chose its 3,000-gene panel across studies and assigned harmonized labels. This is a **cross-acquisition-source test conditional on a shared panel and annotation**, not a blind independent-label validation, donor-level replication, drug-response measure, or clinical result. These are neural **organoids**, not perfused organ-on-chip cultures; the score does not validate cell typing in a chip. Source-internal random-cell and held-bio_sample validation scored 0.953046 and 0.955731 on different validation rows. Their differences from the cross-study score describe protocol changes and do not isolate the cause of a performance gap.

The same evaluation discipline is illustrated on 3,072 expert-labeled organ-on-chip brightfield

images [1, 2]. The source ZIP has 57 of 57 date-like test prefixes also present in training. A frozen image model scored 0.799925 balanced accuracy on its 656-image source test and 0.657920 on a different 736-image prefix-held-out test. On the 151 shared images, the paired difference was +0.066088 with a 14-prefix interval of -0.042091 to +0.122483. That interval crosses zero; the prefixes are not verified physical chip IDs. These two cases show the same practical rule: publish source and group denominators, false alarms, comparable test rows, and uncertainty beside the headline score. The workflow uses public data and free CPU computation, without a paid API or proprietary model.

Keywords: neural organoid; single-cell phenotype; cross-study validation; organ-on-chip; evaluation audit; grouped bootstrap; reproducibility.

2 Research problem and intended use

2.1 Two decisions at the bench

A neural-organoid team may want to check NPC, Neuron, and Glioblast cell-type calls across studies. Within an atlas-selected cohort, a model trained on another study can rank sample keys for annotation review, but a high aggregate score can conceal rare-class false positives or a few dominant samples. The evaluation must therefore show each class’s errors, the full source split, and uncertainty over sample keys. We do not use model output to replace marker-gene review or the study’s original annotation process; a wholly unannotated batch needs a separate end-to-end test.

An organ-on-chip imaging team may train a classifier to send low-quality brightfield images for manual review. Before using that model, the team needs to know whether held-out images truly come from acquisition contexts absent from training. It also needs the number of good cultures that would be incorrectly flagged. A single balanced-accuracy value hides both questions.

The Phenotype Transfer Map answers the organoid decision with three explicitly different measurements: source-held-out macro-F1 and sample-key interval, model-score review order within the atlas-selected three-class cohort, and a retrospective known-label error audit. A single aggregate test score does not show that Glioblast precision is 0.679004 or that the highest-scored review keys have 1.721 times the equal-key mean error rate. The map is exploratory evidence for review order, not proof of benefit on wholly unannotated future batches.

The nearest existing work is the source [HNOCA atlas and its tools](#) [9]: its [AtlasMapper](#) already maps query cells, transfers labels, and computes presence scores with a grouping option. Khatri and Bonn previously studied calibrated uncertainty for single-cell label transfer [15]. This report does not claim a new label-transfer algorithm, atlas, or generic uncertainty estimator; our

margin score is not calibrated as an error probability. Its contribution is a reproducible **decision record** on one whole-study holdout: the fixed predictor’s group-resampled transfer result, score-prioritized review lift, and known-label false positives are tied to the same 34 sample keys. Whether that record improves review decisions on an independently labeled, wholly new cohort remains untested.

A separate organ-on-chip image audit answers the imaging decision. Its inputs are one record per image from each evaluation: a stable image ID, a group identifier, the reference good/bad label, the split assignment, and a test prediction. The group might be a physical chip, an experiment, an imaging session, or a weaker proxy. The scientist must identify which it is from the source data. The tool computes what the records support and labels the comparison’s evidence level. It does not infer a physical chip ID from a filename.

Our worked example uses the public image-quality dataset. In it, the first six digits of each image ID resemble a date. We use that prefix to ask whether the source folder split reuses acquisition context across training and test. The prefix is not a verified chip identifier. Accordingly, the user-facing report says “acquisition-context overlap,” not “chip leakage.”

2.2 Why this is an AI-for-life-science problem

The useful unit of generalization depends on how the life-science data were collected. Many cells from the same biological sample are not independent laboratories; many images from the same imaging context are not independent future acquisitions. The organoid case has independently collected expression experiments but a shared atlas panel and annotation. The image case has expert quality labels and real brightfield images but no verified physical chip IDs. Reporting those distinctions helps a researcher decide what to check next. Neither dataset supplies toxicity outcomes, neural connectivity labels, or drug-response ground truth; none is claimed here.

The prior classification study using this same source data reports a single image-level split and identifies stability across splits as future work [3]. Our contribution is a different question: expose grouping assumptions, test-set composition, and same-image prediction changes using a reusable record contract. We neither reproduce its InceptionV3 model nor claim superiority over it.

3 Data, provenance, and permission

3.1 Source material

The [HNOCA v1 archive](#) [9, 10] supplies `hnoca_minimal_for_mapping.h5ad`, a 2,880,860,613-byte file whose published MD5 is 078675d6108e93cebc99676b6b0626aa. A clean free Kaggle

CPU run downloaded the entire file and recomputed that MD5. It contains a 1,770,578-by-3,000 sparse count-like expression matrix, harmonized `annot_level_1` labels, publication names, and `bio_sample` keys. We select Velasco 2019 [11] as development and Bhaduri 2020 [12] as a whole-study test. The selected three classes contain 135,053 and 207,871 cells respectively. Their 21 and 34 complete `bio_sample` keys do not overlap. Some composite Bhaduri keys contain the text “Velasco, 2019”; source assignment uses the separate `publication` field, never a name parsed from the key. These keys are not established physical organoid or donor identities. HNOCA’s global highly variable gene choice and harmonized annotation used information from the studies represented in the atlas, including the held-out acquisition source. We do not use its cross-study scPoli model weights or embedding.

For a separate annotation-source sensitivity test, the [official HNOCA cleaned archive](#) holds the same Bhaduri cells and preserves `cell_type_original`; Zenodo lists this 18,740,061,388-byte archive as CC BY 4.0 with published MD5 409a14a9772e6488c90cda7900b474ef. We did **not** download or independently hash the whole cleaned archive: HTTP Range requests retrieved only needed metadata. All 223,453 Bhaduri HNOCA row keys were unique and identical in order between the cleaned and 3,000-gene archives (ordered SHA-256 77a0ff74a31468084253534e0244d3368b706e0bcb3514d067f821f74d5986ee). [HNOCA’s curation code](#) preserved original observation fields; the [Bhaduri sfaira loader](#) constructs `celltype` from the original paper’s supplementary `Type` and `Subtype` columns. These are a second annotation source for the **same** Bhaduri acquisition, not a fresh laboratory or independently blind gold standard.

The [Organ-on-a-Chip Image Dataset](#) [1] contains 3,072 PNG images, a spreadsheet of experimental metadata and expert image-quality labels, and original train/validation/test folders. The accompanying data descriptor explains image capture, cultures, and expert labeling [2]. The original table’s quality code is 1 for good and 2 for bad, as checked against the source ZIP folders. We map good to 0 and bad to 1 in the audit records. Six cell types occur in the table. No patient-level or private clinical data are used.

The spreadsheet is 119,712 bytes and is checked by SHA-256 before parsing. The image ZIP is 6,710,767,405 bytes; its downloaded content matched the Zenodo MD5 8f7e058996203d48eb03b2d86c0a2e4d. Extracted image IDs are checked one-to-one against the 3,072 table rows. The source split is read from the ZIP64 central directory via range requests, then every mapped ID, folder label, and cell type is reconciled to the spreadsheet. The source directory mapping has 2,130 training, 286 validation, and 656 test images. The code stores raw images and derived feature records locally; the public code repository does not redistribute them.

3.2 License boundary

HNOCA’s minimal and cleaned archives have separate official Zenodo CC BY 4.0 records [9, 16]. The organ-on-chip image Zenodo record lists CC BY 4.0 for its files [1], while the associated descriptor refers to the image dataset as CC-BY-SA without a version [2]. We report both image statements. Neither raw dataset nor derived per-cell or per-image records are included in our code repository. Users obtain data from the attributed source. Our own code is MIT licensed. NumPy and SciPy use BSD-3-Clause, Pillow describes its license as MIT-CMU, scikit-learn and h5py use BSD-3-Clause, and Requests uses Apache-2.0 [4-8, 13, 14]. The source of each external component is listed in the references.

3.3 Evaluation groups

For the alternative holdout, define `prefix = image_id[:6]`. To make the split reproducible without choosing groups based on their labels, compute `SHA256("26" + prefix)` as an integer, then take the remainder modulo 100. Buckets 0-19 are test, 20-29 validation, and 30-99 training. This yields 38/6/15 prefixes and 2,084/252/736 images respectively. The grouped test has 376 good and 360 bad images. The original source test has 365 good and 291 bad. All 57 date-like prefixes in the source test also occur in its training data; none of the 15 grouped-test prefixes occurs in the grouped training data.

These group counts refer to filename prefixes. They are evidence of overlap in that observed identifier. They do not count independently known physical chips, and they cannot establish whether different prefixes belong to the same chip.

4 Audit design

4.1 Record contract and validation

The reusable audit takes two JSONL files. Each line has `id`, `group`, `label`, `split`, and `prediction`; an optional `subgroup` can carry a cell type. Labels are 0 for good and 1 for bad. Predictions are 0 or 1 on test images and null on training and validation images. Each evaluation must contain one record per ID. The program rejects duplicate IDs, missing groups, invalid labels or splits, predictions on non-test rows, and conflicting labels or groups for a shared ID. It does not invent a class score when only one class is present.

For each evaluation, it counts images by split and label, computes the test confusion matrix, balanced accuracy, recalls, and good-image false-positive rate, then counts test groups also seen in training. It intersects the two test sets by exact image ID and recomputes the same measures

on the shared images. A four-cell transition table records whether both models were right, only the first was right, only the second was right, or both were wrong.

Balanced accuracy is the arithmetic mean of good and bad recall. With confusion matrix rows as true good/bad and columns as predicted good/bad, write the cells as $[[GG, GB], [BG, BB]]$. Then good recall is $GG/(GG+GB)$, bad recall is $BB/(BG+BB)$, and the good-image false-positive rate is $GB/(GG+GB)$. A missing class makes balanced accuracy undefined rather than zero.

4.2 Paired uncertainty calculation

On the shared test images, we resample observed prefix groups with replacement, using the **same chosen groups** for both models. Each replicate recomputes balanced accuracy on each model’s predictions and stores the source-minus-grouped difference. We take the 2.5th and 97.5th percentiles of 2,000 valid replicates with seed 26. There are 14 prefixes in the shared set. This is an approximate descriptive interval for this dataset and this observed grouping, not a confidence guarantee across physical chips or new laboratories.

4.3 Evidence categories

The HTML and JSON distinguish three claims. Group overlap is directly observable for the supplied group IDs. The full-test score difference compares different images and is descriptive. The shared-test comparison holds test image IDs fixed, but the two fitted models still use different training rows and different validation rows and thresholds. It therefore narrows one source of confounding without isolating the cause of the difference. All conclusions remain specific to one dataset.

5 Cross-study neural-organoid phenotype audit

5.1 Frozen question and sample accounting

Before model fitting, we fixed NPC, Neuron, and Glioblast as the primary classes. A four-class plan was stopped before training because the held-out Bhaduri Astrocyte label had 1,312 cells in only six `bio_sample` keys, with at least 100 cells in only four keys. **This choice inspected target-study label counts before training, so the endpoint definition was not blind to the target study.** The three-class training set contains 22,587 NPC, 94,395 Neuron, and 18,071 Glioblast cells from Velasco (135,053 cells, 21 keys). The whole Bhaduri test contains 110,067 NPC, 94,733 Neuron, and 3,071 Glioblast cells (207,871 cells, 34 keys). NPC and Neuron occur in all 34 test keys; Glioblast occurs in 20, with at least 100 cells in eight. Astrocyte remains descriptive and is not silently included in the reported three-class score.

We fixed the main decision before the full run: a whole-study test macro-F1 above 0.45, and a paired `bio_sample` bootstrap interval for the main-minus-best-prespecified-comparator difference entirely above zero, would justify presenting the model as a review aid. This does not set a biological deployment threshold. The preregistered point prediction for whole-study macro-F1 was 0.55-0.83; the observed 0.907831 exceeded that range. The source-internal random and grouped validation predictions were also lower than the observed scores; none was rewritten to fit the result.

5.2 Count transform, model, and comparison

We extract the same 3,000 HNOCA genes for selected cells. For each cell, we divide its panel counts by that cell’s panel total, multiply by 10,000, then apply `log1p` to nonzero values. This fixed operation does not fit on the test rows, although the **choice of the panel itself was made by HNOCA across studies**. We fit `SGDClassifier` with log loss, L2 penalty, `alpha=1e-4`, 30 maximum iterations, tolerance `1e-3`, balanced class weights, averaged coefficients, one fitting thread, and seed 26. No Bhaduri labels enter the transform definition, weight fitting, threshold selection, or hyperparameter choice. The full model is trained on all Velasco cells and evaluated once on all selected Bhaduri cells.

As source-internal diagnostics, we fit the same specification separately after a seed-26 stratified 20% cell holdout and a seed-26 `bio_sample` holdout of about 20% of keys. The random split trains on 108,042 cells and validates on 27,011; all 21 source keys appear on both sides. The grouped split trains on 96,022 cells in 16 keys and validates on 39,031 cells in five other keys. Because groups have unequal sizes, 20% of keys does not mean 20% of cells. These are **three separately fitted models and three different test rows**. The score gaps are descriptive; they are not a causal leakage estimate.

We fixed two inexpensive external comparators: predict the most common Velasco class for every Bhaduri cell, and assign each transformed Bhaduri cell to the highest-cosine-similarity Velasco class centroid. The centroids use only Velasco rows. We report the larger comparator macro-F1, but this choice uses external test scores and its paired interval is conditional on that selection. No comparator was added or retuned after seeing Bhaduri outcomes.

For uncertainty, we compute a 3-by-3 confusion matrix for each of the 34 Bhaduri `bio_sample` keys. In each of 2,000 seed-26 draws, we sample 34 keys with replacement and sum their entire matrices, recompute three-class macro-F1, and take the 2.5th and 97.5th percentiles. The same sampled keys are used for the main model and selected comparator, so their difference is paired at the sample-key level. All 2,000 draws retained all three true classes. This interval expresses variation over observed keys under this resampling scheme, not over independently established

donors or physical organoids.

5.3 Measured external result and failure cases

Evaluation	Test cells / keys	Macro-F1	Interpretation
Velasco random cell holdout	27,011 / 21 overlapping	0.953046	Source-internal diagnostic
Velasco held bio_sample keys	39,031 / 5 disjoint	0.955731	Source-internal diagnostic
Bhaduri whole-study test	207,871 / 34	0.907831	Cross-acquisition-source test with a shared atlas panel and labels
Velasco majority class on Bhaduri	207,871 / 34	0.208706	Fixed comparator
Velasco nearest centroid on Bhaduri	207,871 / 34	0.806783	Best of the two prespecified comparators

The Bhaduri macro-F1 95% sample-key interval is **0.842324-0.937195**. The main model exceeds the nearest-centroid comparator by **+0.101048** on exactly the same cells, with a paired sample-key interval of **+0.070494 to +0.120536**. Both prespecified screens passed. The grouped source validation score exceeds the whole-study score by 0.047900, while random source validation is 0.002685 *lower* than grouped validation. Thus this dataset does not show a monotone random-split inflation story. Different test compositions, training sets, and shared atlas preprocessing prevent attributing any one gap to sample overlap.

True class / predicted class	NPC	Neuron	Glioblast
NPC (110,067)	106,403	2,370	1,294
Neuron (94,733)	3,207	91,467	59
Glioblast (3,071)	159	50	2,862

NPC, Neuron, and Glioblast F1 scores are 0.968022, 0.969855, and 0.785616. Here **Glioblast is HNOCA’s neural progenitor cell-type label**, not a glioblastoma diagnosis. Glioblast recall is 0.931944, but precision is **0.679004**: 209 true Glioblast cells are missed, while **1,353** NPC/Neuron cells are incorrectly flagged. Of 4,215 predicted Glioblast cells, about 32.1% are

false positives. One Bhaduri key with 3,030 selected cells has eight consensus-labeled Glioblast cells but 96 model calls. A researcher can place that sample ahead of lower-disagreement samples for marker-gene and annotation review. The prediction alone cannot establish that those 88 extra calls are truly of the annotated cell type. No test key was removed after inspection, and the output retains every key’s denominators and matrix.

5.4 Score-only ordering within an atlas-selected cohort

The error audit above uses known HNOCA labels; it cannot rank an unannotated future sample by its observed mistakes. Before analyzing the saved decision scores, we registered a separate ranking: for each cell, take the highest minus second-highest of the three `SGDClassifier.decision_function` values, transform that nonnegative margin to $1/(1+\text{margin})$, then average these values within each `bio_sample` key. The rank uses model scores and keys only. The transform is monotone and is **not** a calibrated probability of being wrong.

For a retrospective check, we compare that ranking with each key’s observed error fraction against HNOCA labels. We give the 34 keys equal weight, compute Spearman rank correlation, and resample keys with replacement for 2,000 seed-26 draws to form a percentile interval. Before reading this result, our working prediction was correlation 0.2-0.6, with a continuation gate of correlation at least 0.35, interval lower bound above zero, and the top eight uncertainty keys having at least 1.3 times the all-key mean error rate. The observed correlation was **0.779068**, interval **0.554570-0.901040**; the top-eight ratio was **1.720951**. All three exploratory gates passed. The prediction interval itself was exceeded, and it was not rewritten after observation.

After observing the top-eight ratio, we separately registered a conditional rate-shuffle check before running it. We hold the score-based top-eight selection fixed and randomly permute the **34 observed group error rates** with seed 26 for 10,000 draws; each draw compares the top-eight equal-key mean rate against the same all-key equal-weight denominator. Zero shuffles reached the observed 1.720951 ratio, giving a one-sided, plus-one Monte Carlo p of **1/10,001 = 0.00009999**. This checks association against random reassignment of these observed rates to these ranks. Because we chose the queue and inspected its ratio first, the p-value is post hoc and does not correct for that choice, group-size differences, shared atlas labels, or future cohorts.

This check reused the already analyzed Bhaduri source. The cells entering its three-class cohort were selected by their HNOCA labels; only the subsequent score-based ordering is label-free. Its interval describes variation over those 34 keys under the stated resampling method; it is not an external confirmation that the queue will work for a wholly unannotated new laboratory, organoid, or label convention. We chose and inspected this one simple score-derived queue after

seeing the classifier’s main external result, so its apparent utility should be treated as exploratory. The 3,030-cell example above ranks sixth by its **known-label false-positive count** but 27th by the **score-only uncertainty rank**. The two lists answer different questions and are displayed separately; neither ranking should silently stand in for the other.

5.5 Original-author annotation sensitivity: comparator reversal

After the harmonized-label analysis, we preregistered a secondary two-class endpoint using the Bhaduri paper’s preserved original **Type** and **Subtype** labels. Within the already HNOCA-selected three-class test cohort, original labels beginning **Excitatory Neuron** or **Inhibitory Neuron** are **Neuron**; those beginning **Radial Glia** are the other class. Labels containing **Low quality/Low-Quality**, or belonging to **Mixed**, **Unknown**, **Astrocyte**, **Endothelial**, or another class, are excluded. This retains **175,160 of 207,871 cells** across all **34 Bhaduri bio_sample** keys: **79,356 Neuron** and **95,804 Radial Glia**. We collapse the fixed model’s **Neuron** prediction to positive and **NPC/Glioblast** to negative. No original-author label enters feature selection, training, classifier tuning, or the cohort selection; HNOCA’s labels still selected that cohort.

On this distinct endpoint, the fixed model reached binary macro-F1 **0.941193**, with a seed-26, 2,000-draw whole-key bootstrap interval **0.922403–0.956609**. The prespecified source-trained nearest-centroid comparator reached **0.947220** on the same retained cells; the paired model-minus-centroid difference was **−0.006027** with interval **−0.010067 to −0.002531**. The Velasco source-majority comparator scored **0.311792**. Thus the model’s 0.941 score cannot be offered as another superiority result: **the nearest-centroid ordering reverses** when the annotation source and class endpoint change. The model confusion matrix, with original-author **Radial Glia/Neuron** rows and predicted non-neuron/**Neuron** columns, is `[[89823,5981],[4247,75109]]`; nearest centroid is `[[91041,4763],[4404,74952]]`. `author_label_sensitivity.json` preserves all 34 paired group matrices, exclusions, interval inputs, and archive provenance.

This is a joint label-convention and two-class-endpoint sensitivity check on the same Bhaduri acquisition, not a second acquisition-source test, fully unselected batch, or independently blind label set. It does not isolate whether the author-label convention, the class collapse, or their interaction causes the comparator reversal. HNOCA curation may have inspected the original-study annotations. The continuation gate for a new model-advantage result required a positive paired interval against nearest centroid; it did not pass. We retain the reversal because a single harmonized-label macro-F1 would otherwise hide that model choice depends on the evaluation endpoint.

The three full Velasco fits emitted scikit-learn’s maximum-iteration **ConvergenceWarning**; we

did not raise the iteration limit after viewing the test. Numerical stability under longer or alternative optimization remains to be measured. The model’s linear decision values are not calibrated probabilities. HNOCA’s unified annotation, cross-study gene selection, and the low number of keys with large Glioblast counts are material limits even though the prespecified score screen passed. Bhaduri’s original-study labels are now matched to these cells, but they are from the **same acquisition** and were available to the atlas curation process; they are not an independently blind reference. This result supports **annotation review prioritization under the HNOCA label convention**, not clinical classification, drug toxicity, neural function, or an unbiased estimate of biological composition.

6 Image features and frozen model protocol

6.1 Low-cost image representation

Each source PNG is converted to grayscale and bilinearly resized to 128 by 96 pixels. The 29-dimensional feature vector contains five intensity percentiles, mean, standard deviation, Laplacian variance, mean horizontal and vertical absolute differences, a thresholded edge fraction, eight normalized histogram bins, four quadrant means, four quadrant standard deviations, and variation of row and column means. Every feature is finite-checked. This representation was selected as a low-cost baseline, not as a claim of state-of-the-art morphology recognition. Source images are only read to compute these local features; the report itself audits evaluation records.

6.2 F1: initial baselines

F1 fits a deterministic L2-regularized logistic classifier on the grouped split. Metadata-only features are cell type, seeding density, time after seeding, day, and flow rate. Missing-value medians and scaling are estimated on training rows only. The image-only model uses the 29 grayscale features; training-only scaling and fitting are followed by a validation-only threshold chosen to maximize balanced accuracy, with fixed tie rules. The constant-good baseline is included to expose the 0.5 balanced-accuracy floor. F1’s image test score was observed before we selected the F2 model, so later reuse of this test is adaptive.

6.3 F2: one prespecified nonlinear comparison

After F1, we registered one random-forest comparison using the same 29 features and grouped split: 300 trees, `min_samples_leaf=8`, `max_features=sqrt`, `class_weight=balanced_subsample`, seed 26, and the remaining scikit-learn defaults. No new features are added. The threshold is selected using validation probabilities only. Candidate thresholds include distinct probabilities,

adjacent midpoints, and 0, 0.5, and 1. The first selection tier requires bad recall at least 0.60 and good recall at least 0.55; within an eligible tier, maximize balanced accuracy, break ties by distance to 0.5, then by higher threshold. The grouped validation set selected 0.314465497255814 in the Linux reference; the earlier macOS run selected 0.31713680975737424.

The continuation gate, written before fitting F2, required test balanced accuracy at least 0.65, bad recall at least 0.60, **and** good-image false-positive rate at most 0.45. F2 passed the first two conditions and missed the third. We stopped optimizing automatic image rejection on this repeatedly viewed test.

6.4 F3 and F4: split comparison without model search

F3 uses the same feature vector, forest configuration, and threshold rule, now with the original ZIP’s train/validation/test folders. Its validation threshold is 0.503368900036556 in the Linux reference; the earlier macOS run selected 0.4984234764442013. F4 refits both frozen configurations and scores them on the intersection of their test image IDs. F4 changes no feature, model parameter, or threshold rule. The source and grouped full-test results were already seen when F4 was chosen; F4 is a diagnostic paired comparison, not a fresh blinded validation. A second dataset would be required to test transport beyond this source.

7 Measured results

7.1 F1 and F2 on prefix-group holdout

Evaluation	Test good/bad	Balanced accuracy	Bad recall	Good images flagged bad
Always good	376/360	0.500000	0.000000	0/376
Metadata	376/360	0.523345	0.472222	160/376
logistic				
F1 image logistic	376/360	0.608836	0.802778	220/376
F2 image forest	376/360	0.657920	0.797222	181/376

Confusion matrices (true good/bad rows, predicted good/bad columns) are: always good $[[376, 0], [360, 0]]$; metadata $[[216, 160], [190, 170]]$; F1 image $[[156, 220], [71, 289]]$; and Linux F2 forest $[[195, 181], [73, 287]]$. The F2 validation set has 179 good and 73 bad images, matrix $[[99, 80], [19, 54]]$, and balanced accuracy 0.646399.

F1 improves balanced accuracy by 0.085491 over metadata, but falsely flags 58.51% of good test

images. Linux F2 lowers that count to 181 and improves balanced accuracy by 0.049084, while its good-image false-positive rate remains $181/376 = 0.481383$. This exceeds the prespecified 0.45 ceiling, so the automatic quality-gate direction was stopped. A 2,000-resample bootstrap over its 15 test prefixes gives an approximate balanced-accuracy interval of 0.575-0.730; these prefixes are not verified chips. F2 reused a test already viewed for F1 and cannot be called a fresh blind evaluation.

7.2 Full tests under different splits

Evaluation	Test	Balanced	Good-image		
	good/bad	accuracy	Good recall	Bad recall	false positives
F3, source folders	365/291	0.799925	0.843836	0.756014	57/365
F2, prefix groups	376/360	0.657920	0.518617	0.797222	181/376

The Linux source matrix is $[[308, 57], [71, 220]]$; the grouped matrix is $[[195, 181], [73, 287]]$. Source validation has 163 good and 123 bad images, matrix $[[135, 28], [32, 91]]$, and balanced accuracy 0.784029. A 2,000-resample bootstrap over source test prefixes gives an approximate balanced-accuracy interval of 0.744-0.848.

The source-folder test has 57/57 date-like prefixes also present in source training. The grouped test has 0/15 prefixes in grouped training. The Linux full-test balanced-accuracy gap is 0.142005. However, the tests differ in composition: 505 source-test images are absent from grouped test, and 585 grouped-test images are absent from source test. The two training sets, validation sets, and selected thresholds also differ. The gap is an audit signal, not a causal estimate of the cost of overlap.

7.3 Same 151 images under both models

The test intersection has 68 good and 83 bad images from 14 date-like prefixes. It is smaller than either full test. The table below reuses those exact IDs on both sides.

Model	Balanced	Good-image false		
	accuracy	Good recall	Bad recall	positives
F3 source-split model	0.756467	0.838235	0.674699	11/68

Model	Balanced	Good-image false		
	accuracy	Good recall	Bad recall	positives
F2 grouped-split model	0.690379	0.573529	0.807229	29/68

The source model’s shared-image matrix is $[[57,11],[27,56]]$; the grouped model’s is $[[39,29],[16,67]]$. The paired balanced-accuracy difference is $+0.066088$. Both models are correct on 92 images; only the source model is correct on 21; only the grouped model is correct on 14; both are wrong on 24. A paired 14-prefix bootstrap gives an approximate 95% difference interval of -0.042091 to $+0.122483$ (2,000/2,000 valid draws). The prespecified strong-evidence gate required a difference of at least 0.08 and an interval lower bound above zero; neither condition passed. The interval does not establish a general direction of evaluation bias.

Holding test images fixed removes test-composition differences. It does not equalize the rows used to train the forests or the rows used to choose their thresholds. It also cannot turn a date-like prefix into a physical chip ID. The result is not evidence that the same chip was used for training and test.

7.4 Cell-type boundaries

The grouped F2 full test includes A549 (151 good/81 bad), CACO (27/55), HPMEC (187/152), HSAEC (11/24), HUVEC (0/39), and NHBE (0/9). With no good images in the last two cell types, their good recall and balanced accuracy are undefined. The generated JSON retains each cell type’s denominator and confusion matrix; no single cell-type result is promoted as a separate discovery.

8 Software artifact and real workflow

The offline workflow makes group identity and error denominators explicit in two biological settings. Its image-audit core reads two JSONL record sets, validates them, and writes `audit.html` and `audit.json`. The HTML summarizes overlap, test denominators and errors, the shared-image comparison, and interpretation limits. The JSON stores input records, counts, metadata, and input file hashes. The organoid route produces a whole-study three-class score, per-class and per-bio_sample matrices, comparator scores, and sample-key intervals from the HNOCA original. Both routes use local or free CPU computation and do not call a model API or upload research data to a service.

For the image case, the `ooc` command rebuilds test predictions from locally extracted image

features and fixed model specifications. Before export, it checks fresh F2 and F3 results against the refitted models and checks F4’s shared-image counts, confusion matrices, transitions, paired difference, and interval. On a prepared real feature cache, Linux ARM64 reproduced the independent Kaggle Linux x86-64 deterministic F2-F4 fields and exported the report with **status: verified**. This checks the image-reporting path; it is not a second biological dataset or a new external validation result.

A researcher with their own per-image predictions can bypass the example adapter. They supply source and grouped JSONL records, with group IDs that reflect their actual experimental provenance. The generic report then exposes group overlap and both test compositions. If the test sets have no shared IDs, it cannot compute a same-image contrast. If a test has only one class, balanced accuracy is undefined. These are intentional boundaries of the evidence.

The organoid route is `python organoid_phenotype.py` on a free Kaggle Linux CPU with Internet enabled. The dependency versions are pinned in `requirements-organoid.txt`; the command downloads and checks the 2,880,860,613-byte HNOCA v1 file against MD5 078675d6108e93cebc99676b6b0626aa, extracts selected sparse rows, fits the fixed Velasco model, and writes `organoid_audit/audit.html`, `audit.json`, and `groups.csv`. The first whole-study run completed with script exit 0 in 669.356 seconds, with 2,554,363,904 bytes peak resident memory. An independent local verifier read all 207,871 scored cell records and recomputed the 34 group matrices, external macro-F1, and 2,000-draw primary interval. A second **clean run of the final public entry point**, from the original file on Kaggle Linux x86-64 with Python 3.12.13 and four free CPU cores, completed with script exit 0 in **675.147 seconds** and peak resident memory **2,561,519,616 bytes**. It exactly reproduced the first run’s 207,871-cell and 34-key denominators, full confusion matrix, macro-F1 0.9078308669478505, interval [0.8423242754145907, 0.9371954104623934], and paired difference interval [0.07049358560024495, 0.12053559237535703]. Ninety-three compared floating fields had maximum observed difference 0; exact integer and key equality was required and $1e-12$ was the preregistered absolute tolerance for floats. The output HTML is rendered from the JSON written by that same run. The source file is removed after extraction; no raw atlas, per-cell predictions, or fitted weights are published with this repository.

The optional `python organoid_phenotype.py --author-label-sensitivity` runs the same model and adds a byte-range read of the separate cleaned archive’s original-author annotations. It validates the complete Bhaduri row-key digest before scoring the prespecified Neuron/Radial Glia endpoint, then writes `organoid_audit/author_label_sensitivity.json` and the same values into `audit.json` and HTML. The JSON holds each key’s paired 2-by-2 confusion matrices, exclusions, labels, and 2,000-draw group interval. The clean archive’s **published** MD5 is documented, but the byte-range run does not claim to recompute the hash of all 18.74 GB.

The [competition-linked public Kaggle notebook](#) separately ran that exact source hash 72752afaf60ee7922f77b6937bd7095846353d3ff9fcb0471f04a3f57ea538a9 from the original HNOCA file. Its job reached COMPLETE with script exit 0 in **2,641.932 seconds** and **2,554,421,248 bytes** peak resident memory; its `original_verified` stage alone took **2,593.253 seconds** to fetch and hash the source. The downloaded notebook output had all 34 CSV rows and exactly matched the first run’s method, group matrices, external macro-F1, and bootstrap interval under an independent output comparison. The notebook is listed on this competition’s public Code page.

The image source-to-report route is `python run_full_audit.py`. It downloads and verifies both original Zenodo files, extracts the 3,072 image feature vectors, computes F1-F4, and writes the offline HTML, JSON, and `run_evidence.json`. On a local machine its default four-minute budget allows the same command to resume the ZIP download and feature extraction. A clean Kaggle Linux CPU run used `--time-budget 0` to complete the route in one invocation, produced the report, matched the Linux reference, and exited 0. The corresponding macOS run uses the same feature SHA-256 but different forest trees; its report marks the platform difference and the command exits 3.

9 Reliability, limitations, and application value

9.1 What is directly observed

The HNOCA source file’s full MD5 and size, 342,924 selected cells, 21/34 distinct source/test `bio_sample` keys, per-cell predictions, class matrices, source-internal and external scores, and 34-key bootstrap outputs are recorded. The Velasco and Bhaduri experiments were collected by separate research teams [11, 12]. The external metric is above the prespecified screen and its difference from the prespecified comparator has a positive paired interval. The rare Glioblast false-positive count remains visible beside the high macro-F1.

The source folder assignments, the labels, the date-like image prefixes, and the predictions are inspectable. For those supplied identifiers, 57 of 57 source-test prefix groups also occur in source training. The different full-test scores and their denominators are observable, as are the same-image predictions on the 151-image intersection. We supply counts rather than only a single summary score so a researcher can see false alarms on good cultures.

9.2 What is not identified

HNOCA’s cross-study panel selection and harmonized labels limit independence: an acquisition-source holdout is not a fully external annotation gold standard. Its `bio_sample` values do not

prove physical organoid or donor identities. The grouped interval resamples 34 keys, and the rare Glioblast class has at least 100 cells in only eight keys. No prospective deployment or independently blinded annotation test is reported.

A filename prefix is not independently validated as a physical chip, biological replicate, or day of acquisition. Even if the apparent date meaning were exact, images from different dates could share a chip, and images from the same date could come from different chips. The dataset does not resolve that ambiguity. The same-image result cannot isolate group overlap from a change of training data or threshold. The paired interval uses only 14 observed prefixes and crosses zero. No second non-pathogen organ-on-chip *image-quality* dataset with verified per-sample groups and labels has yet produced an effect estimate. Repeated attention to this source test during F1-F4 also limits how much the later numeric comparisons can be treated as confirmatory.

The quality labels assess brightfield image suitability. They do not quantify cell viability, therapeutic effect, toxicity, or clinical outcome. In particular, our false-positive counts are flags on expert-labeled good images, not measured damage to cultures. Any production triage workflow would require prospective testing, a definition of the actual acquisition unit, and user-selected costs for missed bad images versus good images sent to review.

9.3 Intended application

For neural-organoid single-cell work, the score-based queue can prioritize keys **within the already atlas-selected three-class cohort** without using those keys' outcome labels in its ordering. When HNOCA consensus labels are available, a separate error audit puts samples with unusually high predicted-versus-consensus Glioblast proportions into an annotation review queue. The per-key tables identify where marker genes or original cell annotations deserve inspection. A wholly unannotated new batch has not passed this end-to-end test; neither queue is automated release of an unbiased composition estimate.

For organ-on-chip brightfield work, the image audit can serve as an evaluation checklist with executable calculations. When a team has real chip IDs, it can use them as group fields and inspect whether its test chips were truly absent from training. When it has only acquisition sessions or batches, the result must carry that weaker name. The same code can also compare two test protocols on their shared images, so apparent improvements on different test sets are not casually treated as paired effects. This remains useful even when an image model does not clear a deployment gate.

10 Reproducibility and dependency disclosure

10.1 Hardware and software

The image audit runs on CPU with Python 3.12. `requirements.txt` pins NumPy 2.5.3, Pillow 12.3.0, SciPy 1.18.1, and scikit-learn 1.9.1. The initial organoid result used free Kaggle CPU with Python 3.12.13, NumPy 2.0.2, SciPy 1.16.3, scikit-learn 1.6.1, and h5py 3.16.0; its public dependency set and clean-run check are reported with the published organoid entry point. Both cases use no paid service, proprietary model, non-public data, or GPU. The source image ZIP is 6.71 GB and the HNOCA archive is 2.88 GB, so source transfer dominates a clean run. Raw source data, generated features, scored cells, and fitted weights remain local or in temporary Kaggle job outputs; they are not redistributed in this repository. Runtime measurements are specific to their stated environments.

10.2 HNOCA original-file run and output checks

The initial free Kaggle CPU job fetched all 2,880,860,613 bytes from Zenodo, recomputed MD5 078675d6108e93cebc99676b6b0626aa, and extracted a 342,924-by-3,000 sparse matrix with 57,695,927 nonzero entries. Its 669.356-second script runtime included 608.708 seconds for download and hash verification; peak resident memory was 2,554,363,904 bytes on four CPU cores. The job reached **COMPLETE** and the script reported exit 0. The result JSON recorded the three-class full-test matrix and every Bhaduri sample key. Its SHA-256 was 67cbaec484e7fc4dded61b3b82b8d317faab80c34c4f5949a51b8ce92291099f and the compressed 207,871-row prediction file's SHA-256 was 37bd891dea647dbcdc77b428643ce8cdeebebe82c908c0db9. A separate verifier read that file and reconstructed the full and per-key confusion matrices, macro-F1 0.9078308669, and 95% sample-key interval 0.8423242754-0.9371954105 in 2,000 effective draws. The comparator predictions and paired interval come from the same full job; the separate verifier did not independently refit or reconstruct the nearest-centroid predictions. We keep that limit explicit.

10.3 Linux reference and independent clean run

An earlier clean Kaggle Linux x86-64 CPU run started without a prepared cache, verified the 6,710,767,405-byte ZIP against source MD5 8f7e058996203d48eb03b2d86c0a2e4d, verified the datasheet SHA-256, and extracted features for all 3,072 image IDs. Its feature-cache SHA-256 was 65bb227546f96b7f4643ccf5f98caf50ca38b90c23cce221367dae7d5c6e5902. That run exited 3 because the reference still contained earlier macOS measurements. Its actual Linux F2-F4 output is now the frozen Linux reference; the macOS record remains separately identified in the same file.

A second Kaggle run used the public one-command code and a new, empty `/kaggle/temp/ooc_audit_cache`. It downloaded the original files again, verified the same source and feature hashes, recomputed F1-F4, and produced `audit.html`, `audit.json`, and `run_evidence.json` with `status: verified`. The private verification job completed with exit 0 on four free Linux x86-64 CPU cores, with GPU disabled and internet access for Zenodo. The audit command took 601.00 seconds and recorded 218.98 MiB peak resident memory. The two report SHA-256 digests are recorded in `run_evidence.json` and match the downloaded artifacts. Its source test had 57/57 date-like prefixes also in training, and the shared test had 151 images, 68 good and 83 bad, from 14 prefixes.

Measure	Clean Linux reference	Original macOS measurement
Grouped F2 test balanced accuracy	0.657920	0.657801
Grouped F2 good images falsely flagged	181/376	179/376
Source F3 test balanced accuracy	0.799925	0.797882
Source F3 good images falsely flagged	57/365	61/365
Same-151-image paired balanced-accuracy difference	+0.066088	+0.070783
Paired 14-prefix bootstrap 95% interval	-0.042091 to +0.122483	-0.031747 to +0.127527

We located the first measured divergence before validation and threshold selection. On the same ordered feature and label arrays, with identical seeds and one fitting thread, the first forest tree is identical across macOS Python 3.12 and Linux x86-64/ARM64 Python 3.12. In a 300-tree grouped fit, tree 3 first differs at node index 96: macOS selects feature 7 at threshold 0.0150570632, while Linux selects feature 9 at 0.0283526825. A 64-row probe first differs in tree 16. Linux x86-64 and ARM64 agree on every probed tree, validation probability, and threshold. The precise lower-level cause of the tree split is not established; the identical source and feature hashes rule out a different image cache as its cause.

The Linux ARM64 check reproduced 820 deterministic reference leaf fields, including 207 floating-point fields, with maximum observed difference zero. The frozen comparison allows only $1e-12$ absolute numeric difference and requires exact counts, labels, split summaries, thresholds' nonnu-

meric selection fields, and confusion matrices. The feature SHA-256 must match exactly; source ZIP MD5 and datasheet SHA-256 are verified before evaluation. This tolerance is far below the measured cross-platform score and threshold differences. On macOS, the report explicitly marks matching features with platform-dependent forest scores and exits 3. A changed source or feature file likewise cannot receive **verified** status.

The grouped false-positive rate exceeds the prespecified 45% ceiling on both systems. The descriptive full-test gap exceeds the audit-direction screen of 0.10 on both systems. The paired difference is below the 0.08 stronger-evidence screen and its interval crosses zero on both systems. We do not overwrite the earlier measurements or infer a general physical-chip leakage effect from either run.

10.4 Reproduction sequence

From the public repository root, create a Python 3.12 virtual environment, install `requirements.txt`, and run `python run_full_audit.py`. The default local time budget pauses with a resumable instruction; repeat the same command until it completes. On a remote CPU without a short process limit, use `python run_full_audit.py --time-budget 0`. The script performs the spreadsheet and ZIP checks, feature extraction, F1-F4 calculations, Linux-reference comparison, and report export. The source mapper reads ZIP64 metadata, reconciles 3,072 IDs and source-folder labels to the spreadsheet, and writes the mapping locally. F4 refits both models and checks the stored F2/F3 results before the offline report is written. The JSON records input SHA-256 hashes and per-image predictions. No private database, model file, paid API, or GPU is needed.

A separate generic route accepts two user-provided record files:

```
python audit_report.py records \  
  --source-records SOURCE.jsonl \  
  --grouped-records GROUPED.jsonl \  
  --output OUTPUT_DIR
```

Each JSONL record must use a stable `id`, a provenance-backed `group`, `label` 0 or 1, `split` train/val/test, and a `prediction` on test only. The record contract lets another team run the audit without adopting our image features or random forest. Their group labels, reference labels, and predictions remain their responsibility.

11 Prior work, originality, and responsible disclosure

The HNOCA atlas, its 3,000-gene panel, harmonized labels, and the original Velasco and Bhaduri experiments predate this competition [9-12]. We did not create those cells or annotations and do not present their label convention as an independent gold standard. The image dataset, expert quality labels, and a same-data classification study also predate this work [1-3]. Our competition work is the fixed cross-study cell-type model and sample-level evaluation, plus the organ-on-chip image split and shared-image audit, offline report routes, and reproducibility checks. All reported numeric results are derived from publicly obtainable inputs by the described scripts. No result or demo is simulated.

The source package’s split was accepted as an object of study, not as an independent ground truth about physical chips. A finding of 57/57 prefix overlap is not a finding of chip-level data leakage. The full-test accuracy difference does not compare identical samples. The paired difference and its crossing-zero interval do not warrant a claim that source splits are systematically optimistic. We make these distinctions because the competition asks for verifiable results and because an evaluation tool loses value if its own headline overstates the evidence.

There is one individual entrant, Yan Su. Python, NumPy, SciPy, scikit-learn, h5py, Requests, and Pillow provide the runtime; no foundation model or paid inference API is part of the quantitative result. AI coding assistance may be used to write and review code and prose, but reported numbers come from real runs and are checked against scored records. Research data remain attributed to their sources and are not redistributed in this repository. The code is released under MIT.

12 References

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13 Appendix A: denominators and decision gates

Version	Test unit / denominator	Primary observation	Decision
F1	15 held-out date-like prefixes, 736 images	Image BA 0.608836; 220/376 good images flagged bad	One prespecified nonlinear comparison allowed
F2	Same grouped test, 736 images	Linux forest BA 0.657920; 181/376 good images flagged bad	Automatic quality gate stopped because 48.14% > 45%
F3	Original folder test, 57 overlapping prefixes, 656 images	Linux forest BA 0.799925; nonpaired gap +0.142005	Audit direction explored; paired check required

Version	Test unit / denominator	Primary observation	Decision
F4	Exact common test IDs, 151 images, 14 prefixes	Linux paired gap +0.066088; interval crosses zero	Strong-evidence gate failed; report as diagnostic
F7	Real indexed images	Offline HTML and JSON with per-image records	One-command report entrance accepted

The F3 candidate screen required source-test balanced accuracy at least 0.75 and full-test gap at least 0.10, both of which were observed on macOS and Linux. That screen only justified the F4 diagnostic, not a leakage claim. The F4 strong-evidence screen required a paired gap at least 0.08 and an approximate interval lower bound above zero; neither was observed on either platform. These thresholds were used as research decisions, not post hoc significance tests.

14 Appendix B: numeric audit checks

The following equalities can be checked against the generated Linux JSON and code:

- F2 grouped full test: $195 + 181 = 376$ expert-good images; $73 + 287 = 360$ expert-bad images. Balanced accuracy is $(195/376 + 287/360)/2 = 0.6579196217$.
- F3 source full test: $308 + 57 = 365$ expert-good images; $71 + 220 = 291$ expert-bad images. Balanced accuracy is $(308/365 + 220/291)/2 = 0.7999246811$.
- Shared test: 68 expert-good plus 83 expert-bad images equals 151 IDs. The F3 matrix is $[[57, 11], [27, 56]]$; the F2 matrix is $[[39, 29], [16, 67]]$.
- Shared-image F3 balanced accuracy is $(57/68 + 56/83)/2 = 0.7564670446$; F2 is $(39/68 + 67/83)/2 = 0.6903791637$; their difference is 0.0660878809.
- Correctness transitions sum to the common denominator: 92 both correct + 21 source-only correct + 14 grouped-only correct + 24 both wrong = 151.
- Test-composition accounting: 656 source test - 151 common = 505 source-only; 736 grouped test - 151 common = 585 grouped-only.

The audit program computes these numbers from records. The appendix shows how to check them by hand; it is not a separate source of values. Full precision, source hashes, and per-image assignments are held in the generated local `audit.json`. The HTML rounds for reading, and the JSON preserves the computed values.